Vision-Guided Cable Connector Insertion with a Kinova Gen3 Lite

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I. Introduction

A. Task and Approach

Plugging in a cable is easy for a person but hard for a robot. It has to find the connector, grasp it without disturbing the cable, line it up with the socket, and push it in despite tight clearances and a plug that can jam. This project automates that sequence with a six-DoF Kinova Gen3 Lite, a two-finger parallel gripper, and a wrist-mounted Intel RealSense D405 depth camera (eye-in-hand). Connectors sit in a 3D-printed cradle at a known place on the table, and their sockets are mounted on a fixed board inspired by the NIST Assembly Task Board 1, using real sockets so that clearances and contact are realistic (Fig. 1).

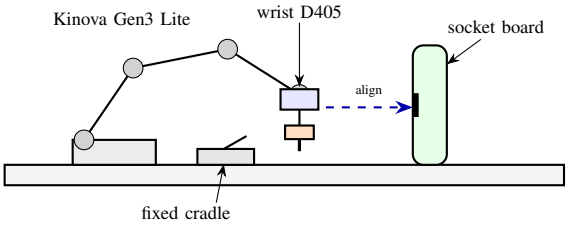


Fig. 1. Schematic of the eye-in-hand configuration, fixed cradle, and marked socket board (conceptual, not to scale).

Three connectors of increasing difficulty are used: a **6.35 mm audio jack** (cylindrical, least sensitive to rotation), an **IEC power connector** (keyed, three pins), and **USB-A** (flat, keyed, tightest clearance); Fig. 2 sketches their profiles.

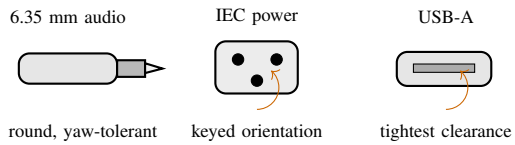


Fig. 2. Conceptual connector profiles and their increasing alignment demands (not dimensioned).

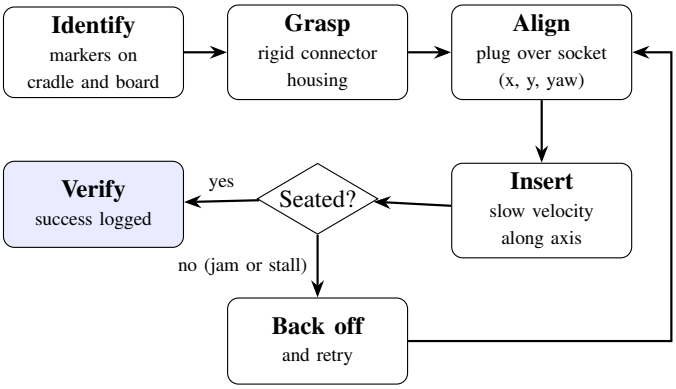


Fig. 3. The five-step pipeline run for every cable. A jam during insertion sends the arm back to re-align instead of ending the attempt.

Fig. 3 shows the sequence. The robot detects markers on the cradle and board, grasps the connector by its rigid housing, moves above the socket, and corrects planar position and yaw from visual feedback. It then pushes the plug in slowly along the insertion axis and checks that it is seated. After a jam or stall, it backs off and realigns instead of ending the trial.

B. Why It Matters

About 542,000 industrial robots were installed worldwide in 2024, and the electronics industry took 24% of them [1]. Yet wiring-harness assembly, which involves handling cables and plugging in connectors, is still largely manual, in some cases up to 90% [2]. The cable pulls on the plug, tolerances are tight, and small position or angle errors cause a jam. The same skill applies to data-center cabling, electrical cabinets, and electronics assembly.

C. Constraints

- **Parallel gripper:** cannot reorient the plug in its fingers; a shift in the grasp or at first contact changes the alignment.
- **Single wrist camera:** the gripper and plug can block the socket near insertion, and there is no global view.
- **No wrist force/torque sensor:** a jam or stall has to be inferred from motion and insertion depth.
- **Tight clearances:** IEC and USB-A are keyed, and small offsets or tilts jam them; USB-A is the hardest.
- **Deformable cable:** its weight and stiffness pull on the plug and add uncertainty.

D. Requirements

The system must:

1) Grasp the rigid housing from the cradle with a stable hold.
2) Locate the socket with the D405 and markers.
3) Align position and yaw, including the keyed orientation, from visual feedback.
4) Insert slowly along the insertion axis to the seating depth.
5) Verify seating and, after a jam, retract, realign, and retry.
6) Demonstrate all three connectors on the physical robot.

E. Prior Information, Design Choices, and Assumptions

Known in advance: connector dimensions and keying (datasheets and NIST CAD files [3]), the cradle pose and its marker, socket positions relative to the board markers (our CAD), D405 intrinsics and hand-eye calibration, and the Gen3 Lite URDF and MoveIt configuration. **Not known:** the exact board pose and the small grasp errors, which vision has to supply.

Design decisions:

- *Grasp the rigid housing, never the cable.* Grasping cable and connector together leads to wrong plug poses and failed insertions [4].
- *Fix the cable in a cradle* so every grasp is nearly the same.
- *Markers* on cradle and board give a 6-DoF pose from one camera and are more reliable than detecting dark sockets [4].
- *Two phases:* MoveIt position control to a standoff pose a few centimeters above the socket, then slow velocity control along the insertion axis.
- *Success from depth and stall:* if commanded motion stops before the expected depth, treat it as a jam, back off, and retry.

Assumptions: the board is static, flat, and reachable; calibration is accurate enough that vision only corrects small errors; and the markers stay visible until the final approach under normal indoor lighting.

F. Target Performance

Each connector is tested in 20 trials. A trial ends when the plug is seated or after three insertion attempts. A success needs the expected depth, so a plug resting in the entrance does not count. Following the NIST task-board style [5], success and time are logged for every trial, and each failure is reported by cause (perception, grasp, or insertion). Targets (Table I) are lower for harder connectors; Lian *et al.* reached 96 to 100% on USB, RJ45, D-sub, and waterproof connectors with industrial force and vision equipment [4], which one wrist camera and no force sensor is not expected to match.

II. Literature Review

Prior work on connector insertion falls into three groups: vision-based alignment, contact and force-based insertion, and benchmarks and learning-based methods. For each paper we note what was done, what could be useful here, and what could limit it, given a low-cost arm with one wrist camera and no force sensor.

A. Visual Servoing and Alignment

Peng et al. [6] compared position-based (PBVS) and image-based (IBVS) servoing on an ABB arm placing a large panel, using markers and two cameras (Fig. 4). Both reached sub-millimeter accuracy in 12 trials each (PBVS 33.9 s, IBVS 45.8 s), and IBVS was more robust only when the desired image was known beforehand. *Useful:* the same ingredients as ours (eye-in-hand, markers, hand-eye calibration) and the PBVS versus IBVS choice. *Limit:* large panels, and force control in the final motion.

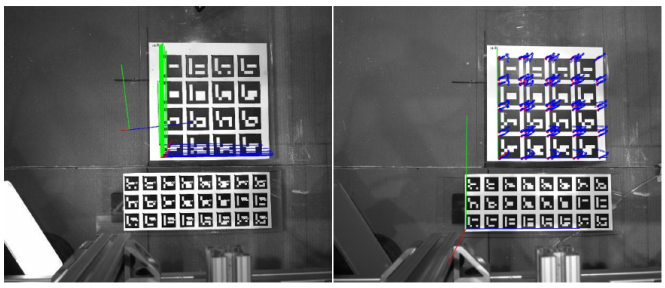


Fig. 4. Gripper-camera views during PBVS (left) and IBVS (right) in Peng et al. From [6].

Haugaard et al. [7] servo a peg onto a hole using a U-Net, trained on 1000 synthetic images, that finds the peg and hole centers in two camera images (Fig. 5). The error is measured between peg and hole, so grasp and calibration errors cancel. The peg is then lowered and finished with compliant force control. It reached 100% success on four holes (100 runs each, 1.6 to 3.6 s), against at most 34% for random search. *Useful:* measuring the peg-to-hole error directly. *Limit:* cameras on a separate robot, position only, force-based insertion.

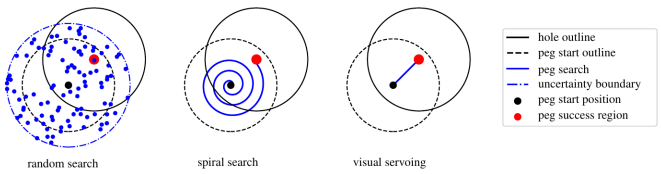


Fig. 5. Random search, spiral search, and visual servoing in Haugaard et al. From [7].

Shen et al. [8] insert a D-sub connector (0.2 mm clearance) from NIST Board 1 with a wrist-mounted D435 (Fig. 6). A network predicts two landmarks on the peg edge, the robot servos them onto the hole to correct x, y, and yaw, descends, runs a small spiral until a force peak shows the peg dropped in, and finishes with constant force. Trained on 3000 synthetic and 420 real images, it succeeded 10/10 times (ORB baseline 8/10). *Useful:* wrist-camera landmark alignment, which fiducial corners could also provide. *Limit:* force peak for insertion, one connector, 10 trials.

TABLE I
PERFORMANCE METRICS, HOW EACH IS COMPUTED, AND TARGET VALUES

Metric	Definition and measurement	Target
Insertion success rate S	$S = n_{\text{seated}}/N$ for each connector, where a trial counts as seated if the plug reaches its expected seating depth d_{seat} and the LED is on; reported with a 95% Wilson confidence interval	Audio jack $\geq$ 90%, IEC $\geq$ 80%, USB-A $\geq$ 70%
First-attempt success S_1	Fraction of trials seated on the first attempt, without any back-off	Reported
Grasp success rate G	Grasps in which the plug stays held after a 10 cm lift and a 3 s hold with no visible slip, divided by all grasp attempts	$\geq$ 95%
Socket localization error	Position error $e_p = \|\hat{\mathbf{p}} - \mathbf{p}_{\text{ref}}\|$ and yaw error $e_\psi = \|\hat{\psi} - \psi_{\text{ref}}\|$ of the estimated socket pose against a reference from the board CAD, checked by a physical measurement; mean and 95th percentile	Mean $e_p < 2$ mm, mean $e_\psi < 3^\circ$
Cycle time T	Time from the start of the pick motion to the LED turning on, over successful trials; mean and standard deviation	Mean < 60 s
Retries per success R	Mean number of back-off-and-retry cycles per successful trial	≤ 1 on average
Failure causes	Share of failed trials caused by perception, grasp, or insertion, assigned from the logged data and video	Reported

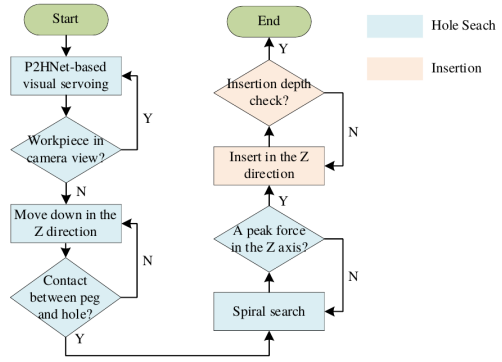


Fig. 6. Insertion strategy of Shen et al.: vision aligns the peg, and a force peak signals insertion. From [8].

Zhang et al. [9] estimate the full 6D pose of an RJ45 or HDMI socket once (Fig. 7): a YOLOv5x detector, plane fitting on depth for roll and pitch, and image warping for yaw. A KUKA arm then moves to a pose above the socket and inserts with impedance control. Position error was about 2 mm and 1.6°, with 94% success for RJ45 and 77% for HDMI. *Useful:* one depth-based pose estimate and a straight approach; the tighter port fails more often. *Limit:* no visual feedback during the push, and impedance control.

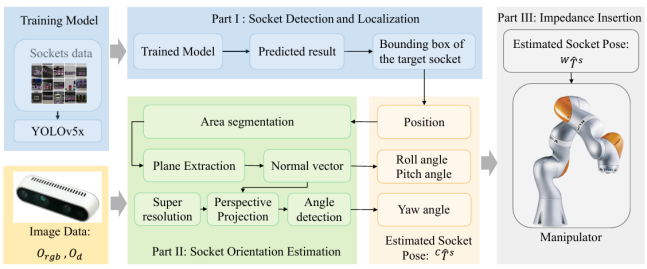


Fig. 7. Pipeline of Zhang et al.: detection, 6D pose from depth, and impedance-controlled insertion. From [9].

B. Contact and Force-Based Insertion

Song et al. [10]: marker-based visual servoing, then a tilt strategy with impedance control, for cable connectors presented by a feeder in a fixed orientation. *Useful:* a repeatable plug presentation, like our cradle. *Limit:* needs a force/torque sensor.

Chen et al. [11]: four force-based search strategies that diagnose and recover from position errors in connector mating. *Useful:* a limited number of recovery attempts. *Limit:* needs a force signal.

Hartisch and Haninger [12]: 3D-printed compliant fingers, soft sideways and stiff along insertion, with a tilted first contact and no force feedback. They tolerated 7.5 mm misalignment, with 2 failures in 84 insertions. *Useful:* fixed plug presentation and tilted contact. *Limit:* fingers tuned to specific connectors.

C. Benchmarks and Learning-Based Methods

Lian et al. [4] tested industrial solutions on 12 NIST Board 1 tasks (Fig. 8). A 3D-vision method was the most reliable (100% except 4 RJ45 failures in 150 trials per task); an end-to-end system reached 96 to 100% on USB, RJ45, waterproof, and D-sub. Failures came from a translucent RJ45, a black USB socket, and cable grasped with the connector. *Useful:* the same connectors, and failures reported by cause. *Limit:* industrial sensors costing thousands of dollars.

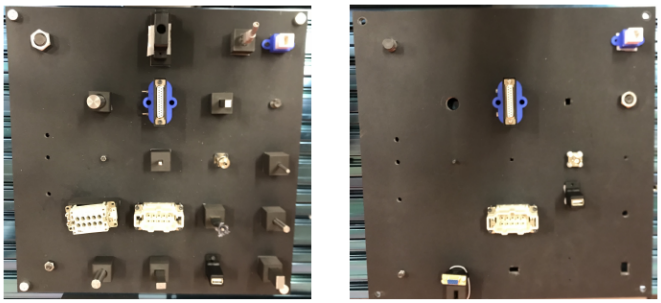


Fig. 8. Nest board and assembly board of Lian et al. From [4].

Kimble et al. [5] define the NIST Assembly Task Boards and a scoring protocol (completion time and a reliability measure from the number of trials). Board 3 includes an audio-jack cable. *Useful:* the protocol and the point that few trials limit how far a success rate can be trusted. *Limit:* cables stay attached to the board.

Schoettler et al. [13] learned USB, D-sub, and Model-E insertion on a real robot from a 32×32 image and a goal-image or electrical-connection reward; residual RL worked best. *Useful:* an electrical signal as an automatic success check. *Limit:* training per connector on the real robot.

She et al. [14] follow a cable with tactile (GelSight) grippers and no camera, and insert a headphone plug into a phone jack (Fig. 9): 98.2% of the cable followed, and 85% insertion success in 20 trials. *Useful:* the audio jack and finding the plug by following the cable. *Limit:* special tactile hardware and one demonstration.

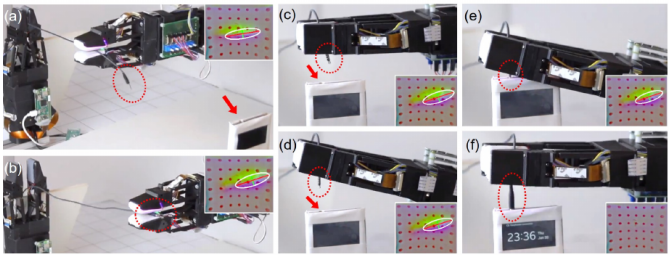


Fig. 9. Cable following and headphone-plug insertion in She et al. From [14].

D. Implications for the Proposed System

The literature suggests a staged system: repeatable pickup, vision-guided alignment, low-speed insertion, and bounded recovery. Our initial controller will use a fixed grasp pose, forward kinematics, hand–eye calibration, and a marker-based socket reference. IBVS (image-feature error) and PBVS (reconstructed pose error) are treated as two separate options, not interchangeable ones.

Without a wrist F/T sensor, we can propose position- and velocity-limited insertion with progress monitoring and a bounded retreat, but cannot claim force-regulated impedance control or reliable contact-state detection. Depth alone cannot prove seating, so success should be checked independently. Our contribution is an experimentally evaluated, low-instrumentation implementation for several connectors, not a claim that visual servoing or search-based insertion is new.

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